BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. Choose the correct Answer for each from the given question:

Q1. These type of waves cannot travel through a vacuum:

(A) Sound waves
(B) Infrared waves
(C) Microwaves
(D) X-rays

Q2. The main source of energy in the stars is:

(A) chemical reaction
(B) nuclear fission
(C) nuclear fusion
(D) Mechanical energy

Q3. The turns ratio of a transformer is 10, it means:

(A) Is = 10Ip
(B) Np = 10 Ns
(C) Ns = 10Np
(D) vs = vp

Q4. The Boolean expression of an AND gate is:

(A) A.B
(B) A + B
(C) A × B
(D) None of these

Q5. Branch of Physics which deals with the charges at rest is called:

(A) Electricity
(B) Electrostatic
(C) Quantum
(D) Magnetism

Q6. ICT stands for:

(A) Integrated Computer Theories
(B) Integrated Circular Technology
(C) Intensive Computer Techniques
(D) Information and Communications Technology

Q7. V^2/R is the formula to find:

(A) current
(B) power
(C) energy
(D) charge

Q8. The presences of dielectric between the plates of capacitors the capacitance of capacitor:

(A) remain constant
(B) decreases
(C) increases
(D) remain unchanged

Q9. The focal length of convex lens is 20cm. The power of lens is:

(A) 4 diopter
(B) 5 diopter
(C) 10 diopter
(D) none of these

Q10. The colour that least deviated by a prism is:

(A) yellow
(B) green
(C) red
(D) violet

Q11. Oscillations are damped due to the presence of:

(A) Linear motion
(B) Restoring force
(C) Fractional force
(D) Mechanical force

Q12. Consider isotope $_{92}U^{235}$ of Uranium. The number of neutrons in it is:

(A) 92
(B) 143
(C) 145
(D) 235

SECTION "B" (Short Answer Question)

Maximum Marks: 24

> Note: NOTE: Answer any EIGHT questions from this section. All questions carry equal marks.

Q2. Write down three characteristics of electromagnetic waves.

Q3. Write one use of each of the following devices:

(i) RADAR

(ii) SONAR

(iii) LIDAR

Q4. Define wave length and derive formula $\lambda = V \times T$

Q5. Explain with diagram, How to change the magnetic field which causes induced emf in a circuit.

Q6. Define thermionic emission. How emission take place in tungsten filament?

Q7. State the heat dissipation law and derive the equation: $H = I^2Rt$

Q8. The wave moving in the pond have a water length 1.6m, and frequency of 0.80Hz. Calculate the speed of these water wave

Q9. Calculate the speed of sound in air at 30°C. Given that speed of sound at 0°C is $331ms^{-1}$.

Q10. A convex mirror used for rear view on a automobile has a radius of curvature 4.00m. If a bus is located at 5.00 m from the mirror. Find the image position and nature.

Q11. Two point charges $q_1 = 5\mu c$ and $q_2 = 3\mu c$ are placed at a distance of 5cm. What will be the Coulomb's force between them?

Q12. Ruby laser emits the beam of red light having a wavelength 694.3 nm. Calculate its frequency.

Q13. A transformer is needed to convert a main supply 220V supply into a 12V supply. If there are 2200 turns on the primary coil, then find the turns on the secondary coil.

Section C (Detailed –Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any FOUR questions from this Section. Each question cries (Six) 6 marks.

Q14. What are short sightedness and long sightedness? How can these defects be corrected with the help of ray diagrams?

Q15. Write Geiger and Morsden α – scattering experiment and its results in detail.

Q16. What is information technology? Also describe five components of computer based on information system.

Q17. State Coulomb's law and derive the equation: $F = 1/4\pi\varepsilon_0 \cdot q1q2/r^2$

Q18. Define fission and fusion reactions, with their equations, Explain fission chain reaction with the help of diagram.

Q19. With the help of circuit diagram derive the formula for equivalent resistance of parallel combination of circuit. Also write three advantages of series combination.